

Detection of increased choline compounds with proton nuclear magnetic resonance spectroscopy subsequent to malignant transformation of human prosta...

In this study, a panel of normal human prostate cells (HPCs) and tumor cells derived from metastases were studied by $(1)H$ NMR spectroscopy to determine whether the malignant transformation of HPCs results in the elevation of choline compounds. Although an elevated choline signal has been observed previously in clinical studies, the contribution of the different Cho compounds to this elevation, as well as their quantification, has not been established until now. Here we have shown that HPCs derived from metastases exhibit significantly higher phosphocholine as well as glycerophosphocholine levels compared with normal prostate epithelial and stromal cells. Thus the elevation of the choline peak observed clinically in prostate cancer is attributable to an alteration of phospholipid metabolism and not simply to increased cell density, doubling time, or other nonspecific effects. Androgen deprivation of the androgen receptor-positive cell lines resulted in a significant increase of choline compounds after chronic androgen deprivation of the LNCaP cell line and in a decrease of choline compounds after a more acute androgen deprivation of the LAPC-4 cell line. These data strongly support the use of proton magnetic resonance spectroscopic imaging to detect the presence of prostate cancer for diagnosis, to detect response subsequent to androgen ablation therapy, and to detect recurrence.